

COMM051 Project D3: Individual Final Report

Name: Lahari Mullaguru

URN:6843990

Group: 10

1. Collaboration Approach

Our team began by analyzing the assignment requirements and understanding the scope of each deliverable. During our initial discussions, we decided to use a structured, task-specific collaboration strategy. This decision was based on the complexity of the assignment and the need to leverage each team member's strengths effectively. To ensure consistent progress, we allocated two weeks each for Deliverable D1 and D2. The choice to combine in-person meetings with Microsoft Teams sessions allowed us to accommodate everyone's schedules and maintain regular communication.

We opted for in-person meetings for tasks requiring group brainstorming, such as deciding on normalization strategies and designing the BI platform, while online meetings were used for task updates and addressing specific challenges. We agreed to divide the tasks among team members based on their expertise, ensuring that each deliverable had dedicated contributors. Regular progress updates during meetings enabled us to identify and resolve issues collaboratively. Ultimately, this approach of combining task division, frequent communication, and leveraging individual strengths was chosen to ensure the timely and high-quality completion of the assignment.

2. Individuals' Roles in the Group

Rishi: Contributed to the design of 3NF tables and established relationships between them. Validated the migrated data to the new structure and set up all dimensions, including Customer, Order, OrderItemDetails, Product, Variant, ProductGroupVariant, and ProductGroup. Designed and implemented the cube with a STAR schema to enable efficient slice-and-dice operations. Assisted peers in resolving blockers and configured calculations for Total Unfulfilled Orders and the Percentage of Unfulfilled Orders. Configured the Product Group to Product Hierarchy using intermediate tables and named calculations and documented the work with screenshots.

Anaswara: Participated in early design discussions to validate data for alignment with the proposed 3NF schema, which led to improvements in the approach. Assisted in verifying data integrity during migration from the old schema to the new one. Identified how to set up Data Source and Data Source Views early in the semester. Created SQL queries for D2 to cross-validate

cube calculations, including cancelled orders and their percentages. Validated cube data against SQL Server manually to ensure accurate calculations.

Shruthi: Finalized and enforced constraints in the 3NF schema, validating foreign key constraints to prevent invalid data operations. Developed prCreateOrderGroup with validation for the OrderNumber format, added error handling, and implemented concurrency management using transaction blocks. Optimized performance by adding indexes to the OrderItem and Order tables.

Sarumathe: Developed prCreateOrderItem with validation for the OrderItemNumber format, incorporating error handling and concurrency management. Contributed to D2 by implementing the calculation for the Percentage of All Fulfilled Basket Orders.

Varsha: Added a geographical hierarchy to the Customer Dimension and indexed critical key pairs to enhance cube processing performance. Contributed to testing and validating cube performance by running queries and verifying results against expected outputs.

3. My Contribution

I contributed to refining the 3NF schema design and created the new tables based on the finalized design, migrating data into the updated structure using proper primary key and foreign key constraints. Implemented triggers to dynamically calculate Saved Total and Total Line Items during row inserts or deletions. Led the configuration of calculations in the cube and contributed to setting up measures. Worked on key calculations, including Cancelled Orders, Percentage of Cancelled Orders, Percentage of Fulfilled Orders, Out of Stock Orders, and Sales Value of Cancelled Orders. Linked the Time Dimension to the cube to establish the required hierarchy.

4. Design Decisions

- It was decided to normalize the provided database schema to Third Normal Form (3NF) to eliminate redundancy and ensure data integrity. This included carefully analysing dependencies and restructuring tables with appropriate primary and foreign keys. The decision was guided by the need for a robust and scalable database design.
- For Deliverable D2, the group opted for a STAR schema design for the business intelligence cube. This decision allowed for simpler and faster slicing and dicing of data, which aligned with the requirements for efficient analysis and reporting.
- Geographical and product hierarchies were incorporated into the BI cube to enable roll-up and drill-down operations. These hierarchies were added to dimensions such as Customer and Product, facilitating granular and high-level analysis.

- The team decided to calculate fields such as SavedTotal and TotalLineItems dynamically using triggers. This ensured that the values were updated automatically on data inserts or deletions, improving accuracy and reducing manual errors.
- Stored procedures were designed with robust error-handling mechanisms to manage transactional integrity. This decision was crucial for maintaining database consistency during concurrent operations or unexpected failures.
- Indexes were strategically added to key columns in the Order and OrderItem tables to improve query performance. This decision was informed by the need to handle large datasets efficiently.
- A time dimension was linked to the cube to enable analysis across hierarchical time units (day, month, quarter, year). This decision supported comprehensive time-based insights.

5. What I Learnt

This assignment provided valuable learning experiences both in technical skills and collaborative teamwork. From a technical perspective, I gained a deeper understanding of database design principles, particularly normalization to 3NF and its role in ensuring data integrity and reducing redundancy. I also enhanced my skills in creating and managing database triggers, stored procedures, and implementing error handling and transaction control mechanisms. Working on the BI platform exposed me to hierarchical data modelling, and the practical use of Microsoft SQL Server 2018 Analysis Services (SSAS/SSDT) for building cubes, measures, dimensions and performing complex calculations.

On the collaboration side, I learned the importance of clear communication and regular updates in ensuring team alignment. Our team use of meetings and task division highlighted how leveraging individual strengths and working collaboratively can lead to a more efficient and productive workflow. I also learned how to navigate challenges, such as resolving discrepancies in data migration which taught me the value of flexibility and collective problem-solving.

Overall, this assignment was a rewarding experience that strengthened both my technical and interpersonal skills.

Appendix: Evidence of Collaboration

